

Revision Notes for Class 7 Maths Chapter 5 Parallel and Intersecting Lines

Parallel and Intersecting Lines is a fundamental topic in Class 7 Maths. In this chapter, students learn the basic relationships between lines on a plane, including what makes lines parallel, intersecting, or perpendicular. Understanding these ideas is vital as they form the foundation for much of geometry in higher classes, and are seen frequently in real-life examples, such as the edges of a table, railway tracks, and the markings on a blackboard.

Intersecting Lines and Vertically Opposite Angles

When two lines cross each other at a point, they are called intersecting lines. The point at which they meet is called the point of intersection. Whenever two lines intersect, they form four angles at the point where they cross. Among these, the angles that are opposite each other are called vertically opposite angles. These pairs of angles are always equal. For example, if one of the angles formed is 120° , the angle opposite it will also be 120° , and the other two angles will each measure 60° .

- Intersecting lines meet at exactly one point.
- At the intersection, the opposite angles are always equal.
- Adjacent angles at the intersection (called linear pairs) always add up to 180° .

One reason measured angles on paper might not add perfectly is due to errors in drawing or using instruments. In geometry, we assume lines are ideal and have no thickness, but in practice, drawing tools are not perfect. Nonetheless, the core properties remain true.

Perpendicular Lines

Perpendicular lines are lines that intersect at a right angle (90°). So, when two lines cross and all four angles formed at their intersection are equal (each 90°), those lines are said

to be perpendicular to each other. Perpendicular lines are common in everyday life, like the corners of a notebook page or the layout of windows in a building.

Parallel Lines

Parallel lines are pairs of straight lines on a plane that never meet, no matter how far they are extended in either direction. For example, opposite edges of a book or railway tracks running side by side are parallel. Both lines must always lie in the same plane. If lines are not in the same plane, they can't truly be called parallel.

- Parallel lines never intersect.
- They always remain the same distance apart.
- In drawings, parallel lines are often marked with arrow signs.

Paper Folding Activities

By folding sheets of paper in different ways—horizontally, vertically, or along diagonals—you can observe parallel and perpendicular lines as crease marks. For example, after making a horizontal fold, you get a line parallel to the edge of the sheet. Making vertical folds creates perpendicular lines to the horizontal ones. Such activities help in visualizing the properties of lines and angles.

Transversals and Angles Formed

When a line cuts across two or more lines, it is called a transversal. A transversal forms eight angles when it intersects a pair of lines. These angles have special names, such as corresponding angles, alternate angles, and interior angles. Not all eight angles can have different measures; they repeat in pairs because of the way the transversal crosses the lines.

- Four pairs of vertically opposite angles are formed.
- Corresponding angles occupy the same relative position at each intersection.

- Alternate angles are on opposite sides of the transversal but inside the lines.
- Interior angles are on the same side of the transversal.

Corresponding and Alternate Angles

If a transversal crosses two parallel lines, their corresponding angles are always equal. For example, if the upper left angle at one intersection is 60° , then the upper left angle at the other intersection will also be 60° . Similarly, alternate angles (like those on opposite corners inside a pair of parallel lines) are also equal. Knowing these properties helps to check if two lines are parallel by simply measuring or comparing the angles made by a transversal.

Interior angles on the same side of a transversal add up to 180° . For example, if one is 120° , the other will be 60° . If corresponding or alternate angles are not equal, the lines are not parallel.

Constructing Parallel and Perpendicular Lines

You can draw parallel lines using a ruler and a set square by sliding the set square along an existing line and drawing another line. If the angles formed between the transversal and two lines are equal, those lines are parallel. Perpendicular lines can also be drawn using the set square or by using paper folding techniques—creating one crease and then folding another at a right angle.

Example Applications of Concepts

The chapter gives examples that use these properties to calculate unknown angles. For instance, if one angle formed by a transversal with parallel lines is given, other angles can be found using the properties of corresponding, vertically opposite, or interior angles. These calculations show why these properties matter, and how geometry helps in everyday problem-solving.

Quick Recap of Key Points

- Intersecting lines cross at one point and form four angles, with vertically opposite angles equal.
- Perpendicular lines meet at right angles (90°).
- Parallel lines never meet and are always the same distance apart.
- A transversal crossing two lines creates eight angles with special relationships.
- For parallel lines, corresponding and alternate angles are equal, and interior angles on the same side add to 180° .
- Folding paper or using set squares helps visualize and construct these lines and angles.

By understanding these properties and practicing with real-life examples and exercises, you will build strong foundations for more advanced mathematics.